

Topics in Machine Learning

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1 Introduction

This is the introduction.

2 Optimizers

2.1 Gradient Descent

3 Architectures

3.1 Transformers

3.1.1 BERT

3.1.2 GPT

3.1.3 Vision Transformers

3.2 Convolutional Neural Networks

3.2.1 ResNet

3.2.2 U-Net

3.3 Recurrent Neural Networks

3.3.1 LSTM

3.3.2 GRU

3.3.3 Mamba

3.4 Graph Neural Networks

4 Generative Models

We discuss generative models in machine learning.

4.1 GANs

4.2 Diffusion Models

4.3 Auto-regressive Models